

```
/*=====
Football Database Project
Database Systems Final Project

Author : Furkan Taha Ünal
Database: Oracle SQL / SQL Server

Description:
Football management database including teams, players,
matches, goals, sponsors and managers.
=====*/
```

```
/*=====
TABLE: NATION
=====*/
```

```
CREATE TABLE NATION (
    NATION_ID    VARCHAR2(50) PRIMARY KEY,
    NATION_NAME  VARCHAR2(50) NOT NULL
);
```

```
/*=====
TABLE: TEAM
=====*/
```

```
CREATE TABLE TEAM (
    TEAM_ID      NUMBER PRIMARY KEY,
    TEAM_NAME    VARCHAR2(50) NOT NULL,
    TEAM_GROUP   VARCHAR2(5) NOT NULL
);
```

```
/*=====
TABLE: MANAGER
=====*/
```

```
CREATE TABLE MANAGER (
    MANAGER_ID   NUMBER PRIMARY KEY,
    MANAGER_NAME VARCHAR2(50) NOT NULL,
    MANAGER_COST NUMBER,
    NATION_ID    VARCHAR2(50),

    CONSTRAINT FK_MANAGER_NATION
        FOREIGN KEY (NATION_ID)
        REFERENCES NATION (NATION_ID)
);
```

```
/*=====
TABLE: EMPLOYEE
=====*/
```

```
CREATE TABLE EMPLOYEE (
    EMPLOYEE_ID  NUMBER PRIMARY KEY,
    FIRST_NAME   VARCHAR2(50),
    LAST_NAME    VARCHAR2(50),
```

```

SALARY      NUMBER,
COMMISSION  NUMBER,
MANAGER_ID  NUMBER,

CONSTRAINT FK_EMPLOYEE_MANAGER
  FOREIGN KEY (MANAGER_ID)
  REFERENCES MANAGER (MANAGER_ID)
);

```

```

/*=====
TABLE: PLAYER
=====*/

```

```

CREATE TABLE FOOTBALL_PLAYER (
  PLAYER_ID  NUMBER PRIMARY KEY,
  PLAYER_NAME VARCHAR2(50) NOT NULL,

  PRICE      NUMBER DEFAULT 0,

  TEAM_ID    NUMBER,
  NATION_ID  VARCHAR2(50),

  CONSTRAINT FK_PLAYER_TEAM
    FOREIGN KEY (TEAM_ID)
    REFERENCES TEAM (TEAM_ID),

  CONSTRAINT FK_PLAYER_NATION
    FOREIGN KEY (NATION_ID)
    REFERENCES NATION (NATION_ID)
);

```

```

/*=====
TABLE: MATCH
=====*/

```

```

CREATE TABLE MATCH (
  MATCH_ID  NUMBER PRIMARY KEY,

  MATCH_DATE  DATE,

  HOME_TEAM_ID  NUMBER,
  AWAY_TEAM_ID  NUMBER,

  SCORE        VARCHAR2(20),

  CONSTRAINT FK_MATCH_HOME
    FOREIGN KEY (HOME_TEAM_ID)
    REFERENCES TEAM (TEAM_ID),

  CONSTRAINT FK_MATCH_AWAY
    FOREIGN KEY (AWAY_TEAM_ID)
    REFERENCES TEAM (TEAM_ID)
);

```

```

/*=====

```

TABLE: GOAL

=====*/

```
CREATE TABLE GOAL (  
  GOAL_ID      NUMBER PRIMARY KEY,  
  
  PLAYER_ID    NUMBER NOT NULL,  
  
  MATCH_ID     NUMBER NOT NULL,  
  
  CONSTRAINT FK_GOAL_PLAYER  
    FOREIGN KEY (PLAYER_ID)  
      REFERENCES FOOTBALL_PLAYER (PLAYER_ID),  
  
  CONSTRAINT FK_GOAL_MATCH  
    FOREIGN KEY (MATCH_ID)  
      REFERENCES MATCH (MATCH_ID)  
);
```

/*=====

TABLE: TEAM_MANAGER

=====*/

```
CREATE TABLE TEAM_MANAGER (  
  TEAM_MANAGER_ID NUMBER PRIMARY KEY,  
  
  START_DATE      DATE,  
  END_DATE        DATE,  
  
  MANAGER_ID      NUMBER,  
  TEAM_ID          NUMBER,  
  
  CONSTRAINT FK_TM_MANAGER  
    FOREIGN KEY (MANAGER_ID)  
      REFERENCES MANAGER (MANAGER_ID),  
  
  CONSTRAINT FK_TM_TEAM  
    FOREIGN KEY (TEAM_ID)  
      REFERENCES TEAM (TEAM_ID)  
);
```

/*=====

TABLE: SPONSOR

=====*/

```
CREATE TABLE SPONSOR (  
  SPONSOR_ID      NUMBER PRIMARY KEY,  
  
  SPONSOR_NAME     VARCHAR2(50) NOT NULL,  
  
  START_DATE       DATE,  
  
  PRICE            NUMBER,  
  
  TEAM_ID          NUMBER,
```

```
CONSTRAINT FK_SPONSOR_TEAM
  FOREIGN KEY (TEAM_ID)
  REFERENCES TEAM (TEAM_ID)
);
```

```
/*=====
TABLE: MATCH_REQUEST
=====*/
```

```
CREATE TABLE MATCH_REQUEST (
  REQUEST_ID    NUMBER PRIMARY KEY,

  REQUEST_TYPE  VARCHAR2(30),

  MATCH_DATE    DATE,

  TEAM_ID       NUMBER,

  CONSTRAINT FK_REQUEST_TEAM
    FOREIGN KEY (TEAM_ID)
    REFERENCES TEAM (TEAM_ID)
);
```

```
/*=====
Football Database Project
Oracle Procedures

Author : Furkan Taha Ünal
=====*/
```

```
SET SERVEROUTPUT ON;
```

```
-----
-- Procedure 1
-- Increase sponsor budget
-----
```

```
CREATE OR REPLACE PROCEDURE UpdateSponsorPrice
AS
```

```
  SponsorNotFound EXCEPTION;

  v_Count NUMBER;
```

```
BEGIN
```

```
  SELECT COUNT(*)
  INTO v_Count
  FROM SPONSOR
  WHERE UPPER(SPONSOR_NAME) IN ('PLAYSTATION','SAMSUNG');
```

```
  IF v_Count = 0 THEN
    RAISE SponsorNotFound;
  END IF;
```

```
  UPDATE SPONSOR
  SET PRICE = PRICE * 1.10
```

```

WHERE UPPER(SPONSOR_NAME) = 'PLAYSTATION';

UPDATE SPONSOR
SET PRICE = PRICE * 1.20
WHERE UPPER(SPONSOR_NAME) = 'SAMSUNG';

DBMS_OUTPUT.PUT_LINE('Sponsor prices updated successfully.');
```

EXCEPTION

```

WHEN SponsorNotFound THEN

DBMS_OUTPUT.PUT_LINE('No PlayStation or Samsung sponsor found.');
```

END;

/

-- Create Match Request

```

CREATE OR REPLACE PROCEDURE AddMatchRequest(

    p_RequestType VARCHAR2,
    p_MatchDate DATE,
    p_TeamID NUMBER

)

AS

    TeamNotFound EXCEPTION;

    v_Count NUMBER;

    v_RequestID NUMBER;

BEGIN

    SELECT COUNT(*)
    INTO v_Count
    FROM TEAM
    WHERE TEAM_ID = p_TeamID;

    IF v_Count = 0 THEN
        RAISE TeamNotFound;
    END IF;

    SELECT NVL(MAX(REQUEST_ID),0)+1
    INTO v_RequestID
    FROM MATCH_REQUEST;

    INSERT INTO MATCH_REQUEST(

        REQUEST_ID,
        REQUEST_TYPE,
        MATCH_DATE,
        TEAM_ID
```

```

)

VALUES(

    v_RequestID,
    p_RequestType,
    p_MatchDate,
    p_TeamID

);

DBMS_OUTPUT.PUT_LINE('Match request created.');
```

EXCEPTION

```

WHEN TeamNotFound THEN

DBMS_OUTPUT.PUT_LINE('Selected team does not exist.');
```

END;

/

-- Add Bonus To Football Player

```

CREATE OR REPLACE PROCEDURE IncreasePlayerPrice(

    p_PlayerName VARCHAR2,
    p_Bonus NUMBER

)

AS

    PlayerNotFound EXCEPTION;

    v_Count NUMBER;

BEGIN

    SELECT COUNT(*)
    INTO v_Count
    FROM FOOTBALL_PLAYER
    WHERE UPPER(PLAYER_NAME)=UPPER(p_PlayerName);

    IF v_Count=0 THEN
        RAISE PlayerNotFound;
    END IF;

    UPDATE FOOTBALL_PLAYER

    SET PRICE = PRICE + p_Bonus

    WHERE UPPER(PLAYER_NAME)=UPPER(p_PlayerName);

    DBMS_OUTPUT.PUT_LINE(
        p_PlayerName || ' salary updated successfully.'
```

```

);

EXCEPTION

WHEN PlayerNotFound THEN

DBMS_OUTPUT.PUT_LINE('Player not found.');
```

```

END;
/

-- Register Goal
-----
```

CREATE OR REPLACE PROCEDURE RegisterGoal(

```

    p_PlayerID NUMBER,
    p_MatchID NUMBER
)
AS
    v_GoalID NUMBER;

    MatchNotFound EXCEPTION;

BEGIN

    IF NOT EXISTS (
        SELECT 1
        FROM MATCH
        WHERE MATCH_ID = p_MatchID
    ) THEN
        RAISE MatchNotFound;
    END IF;

    SELECT NVL(MAX(GOAL_ID),0)+1

    INTO v_GoalID

    FROM GOAL;

    INSERT INTO GOAL(

        GOAL_ID,
        PLAYER_ID,
        MATCH_ID

    )

    VALUES(

        v_GoalID,
        p_PlayerID,
        p_MatchID

    );

```

```

    DBMS_OUTPUT.PUT_LINE('Goal successfully registered.');
```

EXCEPTION

```

WHEN MatchNotFound THEN

DBMS_OUTPUT.PUT_LINE('Match not found.');
```

END;

/

```

/*=====
    Football Database Project
    Oracle Triggers

    Author : Furkan Taha Ünal
=====*/
```

```

SET SERVEROUTPUT ON;

CREATE OR REPLACE TRIGGER TRG_MANAGER_COST

BEFORE UPDATE OF MANAGER_COST
ON MANAGER

FOR EACH ROW

BEGIN

    DBMS_OUTPUT.PUT_LINE(
        'Manager '
        || :OLD.MANAGER_NAME
        || ' salary changed from '
        || :OLD.MANAGER_COST
        || ' to '
        || :NEW.MANAGER_COST
    );

END;
/
CREATE OR REPLACE TRIGGER TRG_PREVENT_TEAM_DELETE

BEFORE DELETE
ON TEAM

FOR EACH ROW

DECLARE

    v_Count NUMBER;

BEGIN

    SELECT COUNT(*)

    INTO v_Count
```



```
FROM TEAM_MANAGER

WHERE TEAM_ID=:OLD.TEAM_ID

AND END_DATE IS NULL;

IF v_Count>0 THEN

    RAISE_APPLICATION_ERROR(

        -20001,

        'Cannot delete team because it still has an active manager.'

    );

END IF;

END;
/
```

```
CREATE OR REPLACE TRIGGER TRG_SPONSOR_DATE

BEFORE INSERT

ON SPONSOR

FOR EACH ROW

DECLARE

    v_Count NUMBER;

BEGIN

    SELECT COUNT(*)

    INTO v_Count

    FROM SPONSOR

    WHERE TEAM_ID=:NEW.TEAM_ID

    AND START_DATE=:NEW.START_DATE;

    IF v_Count>0 THEN

        RAISE_APPLICATION_ERROR(

            -20002,

            'Sponsor already exists for this team on this date.'

        );

    END IF;

END;
```

```
/
CREATE OR REPLACE TRIGGER TRG_MATCH_PLAYER_BONUS
AFTER INSERT
ON MATCH
FOR EACH ROW
BEGIN
    UPDATE FOOTBALL_PLAYER
    SET PRICE=PRICE*1.10
    WHERE TEAM_ID IN(
        :NEW.HOME_TEAM_ID,
        :NEW.AWAY_TEAM_ID
    );
    DBMS_OUTPUT.PUT_LINE(
        'Players received a 10% bonus.'
    );
END;
/
```

```
CREATE OR ALTER PROCEDURE dbo.GetPlayerGoalCount
    @PlayerName NVARCHAR(50)
AS
BEGIN
    SET NOCOUNT ON;
    SELECT
        FP.PLAYER_NAME,
        COUNT(G.GOAL_ID) AS TOTAL_GOALS
    FROM FOOTBALL_PLAYER FP
    LEFT JOIN GOAL G
        ON FP.PLAYER_ID = G.PLAYER_ID
    WHERE FP.PLAYER_NAME = @PlayerName
```

```

GROUP BY FP.PLAYER_NAME;

END;
GO

CREATE OR ALTER PROCEDURE dbo.UpdateSponsorPrice
AS
BEGIN

    SET NOCOUNT ON;

    UPDATE SPONSOR
    SET PRICE = PRICE * 1.10
    WHERE UPPER(SPONSOR_NAME)='PLAYSTATION';

    UPDATE SPONSOR
    SET PRICE = PRICE * 1.20
    WHERE UPPER(SPONSOR_NAME)='SAMSUNG';

    PRINT 'Sponsor prices updated.';

END;
GO

CREATE OR ALTER PROCEDURE dbo.RegisterGoal

    @PlayerID INT,

    @MatchID INT

AS
BEGIN

    SET NOCOUNT ON;

    IF NOT EXISTS(

        SELECT 1

        FROM MATCH

        WHERE MATCH_ID=@MatchID

    )

    BEGIN

        THROW 50002,

        'Match not found.',

        1;

    END;

    DECLARE @GoalID INT;

```

```
SELECT @GoalID=ISNULL(MAX(GOAL_ID),0)+1

FROM GOAL;

INSERT INTO GOAL(

    GOAL_ID,

    PLAYER_ID,

    MATCH_ID

)

VALUES(

    @GoalID,

    @PlayerID,

    @MatchID

);

END;
GO
```

```
CREATE OR ALTER TRIGGER TRG_ManagerSalaryIncrease

ON EMPLOYEE

AFTER UPDATE

AS
BEGIN

    SET NOCOUNT ON;

    UPDATE E

    SET SALARY = SALARY + 100

    FROM EMPLOYEE E

    INNER JOIN inserted I

        ON E.EMPLOYEE_ID = I.EMPLOYEE_ID

    WHERE SALARY < 5000;

    PRINT 'Employee salary updated.';

END;
GO
```

```
CREATE OR ALTER TRIGGER TRG_PreventTeamDelete

ON TEAM

INSTEAD OF DELETE

AS
BEGIN

    SET NOCOUNT ON;

    IF EXISTS(

        SELECT 1

        FROM TEAM_MANAGER TM

        INNER JOIN deleted D

        ON TM.TEAM_ID=D.TEAM_ID

        WHERE TM.END_DATE IS NULL

    )

    BEGIN

        THROW 50010,

        'This team still has an active manager.',

        1;

    END;

    DELETE FROM TEAM

    WHERE TEAM_ID IN(

        SELECT TEAM_ID

        FROM deleted

    );

END;
GO

CREATE OR ALTER TRIGGER TRG_PreventDuplicateSponsor

ON SPONSOR

INSTEAD OF INSERT

AS
BEGIN

    SET NOCOUNT ON;
```

```
IF EXISTS(
    SELECT 1
    FROM SPONSOR S
    INNER JOIN inserted I
    ON S.TEAM_ID=I.TEAM_ID
    AND S.START_DATE=I.START_DATE
)
BEGIN
    THROW 50020,
    'Sponsor already exists for this team on the selected date.',
    1;
END;

INSERT INTO SPONSOR
SELECT *
FROM inserted;

END;
GO
```

```
CREATE OR ALTER TRIGGER TRG_PlayerBonusAfterMatch
ON MATCH
AFTER INSERT
AS
BEGIN
    SET NOCOUNT ON;

    UPDATE FP
    SET PRICE = PRICE * 1.10
    FROM FOOTBALL_PLAYER FP
    INNER JOIN inserted I
    ON FP.TEAM_ID IN(
        I.HOME_TEAM_ID,
```

I.AWAY_TEAM_ID

);

PRINT 'Players received a 10% value increase.';

END;

GO